

HAIYANG DAO, China

WMO: 545870

Lat: 39.055N

Lon: 123.176E

Elev: 33

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 90-04

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.6	17.5	-4.8	4.3	18.5	-1.1	5.2	21.0	41.4	24.4	35.5	21.7	18.9	0	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	84.6	78.1	82.7	76.8	81.0	75.7	79.8	83.0	78.1	81.2	76.8	79.8	6.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.7	149.4	82.7	77.1	141.4	80.4	75.9	135.5	79.1	43.6	83.3	41.8	81.3	40.5	79.9	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
33.2	28.9	25.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	28.6	32.4	39.4	50.1	58.9	68.1	74.4	76.8	70.8	58.9	45.3	33.1
	DBStd	17.50	6.44	5.89	5.48	5.16	5.57	4.25	3.72	3.10	3.73	5.93	7.87	7.20
	HDD50	2270	665	492	331	63	5	0	0	0	0	10	181	523
	HDD65	5283	1130	912	794	447	202	20	0	0	3	197	590	988
	CDD50	3428	0	0	2	66	279	542	758	830	624	287	40	0
	CDD65	968	0	0	0	0	11	112	293	365	178	9	0	0
	CDH74	4345	0	0	0	0	9	216	1384	2251	480	5	0	0
	CDH80	568	0	0	0	0	0	6	194	329	39	0	0	0
Wind	WSAvg	10.4	12.8	11.3	11.8	11.6	10.0	7.9	8.6	7.1	7.9	10.5	12.7	12.8
Precipitation	PrecAvg	33.5	0.4	0.6	0.8	1.9	3.1	3.9	8.4	7.4	3.3	1.7	1.3	0.6
	PrecMax	45.5	1.0	1.8	3.9	3.8	6.9	7.4	18.8	13.5	7.9	3.2	3.0	1.9
	PrecMin	23.7	0.0	0.1	0.0	0.3	0.8	0.8	2.2	2.3	0.7	0.2	0.2	0.2
	PrecStd	6.1	0.2	0.4	0.7	0.9	1.5	1.7	3.1	3.1	1.7	0.9	0.8	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	48.3	57.6	66.0	75.5	80.8	86.9	87.9	83.5	73.9	63.8	51.6
		MCWB	40.5	45.2	48.8	54.1	62.4	68.9	79.7	80.0	75.7	64.8	56.8	47.8
	2%	DB	41.6	43.8	51.7	62.5	72.5	78.4	84.1	85.4	81.3	70.7	60.5	49.1
		MCWB	38.2	39.0	45.5	51.9	61.1	68.1	77.1	79.4	73.9	63.5	56.0	46.0
	5%	DB	39.6	41.8	48.8	60.2	70.3	76.5	82.0	83.4	79.0	68.5	58.6	46.7
		MCWB	35.9	37.7	42.7	50.9	60.1	67.4	76.3	77.9	71.6	61.9	54.7	43.6
	10%	DB	37.6	40.4	46.6	58.1	67.8	74.7	80.1	81.8	77.1	66.4	56.3	44.2
		MCWB	34.1	36.6	41.2	49.6	59.0	66.7	75.4	76.9	69.8	61.0	52.2	40.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.8	46.4	50.7	58.5	65.5	73.1	81.7	83.1	77.6	67.4	59.3	48.9
		MCDB	43.8	48.4	53.8	64.0	72.4	78.0	85.5	85.8	81.5	71.7	61.8	51.1
	2%	WB	38.9	40.5	46.6	55.2	63.4	71.5	79.3	80.4	75.5	65.2	57.3	46.7
		MCDB	41.2	42.8	50.7	59.8	69.3	75.7	82.7	83.8	79.9	68.9	60.1	48.7
	5%	WB	36.5	38.6	44.1	52.9	61.8	70.1	77.4	79.0	73.5	63.7	55.1	44.0
		MCDB	39.1	41.0	47.7	57.7	67.8	73.7	80.3	82.1	77.4	67.2	58.4	46.5
	10%	WB	34.3	37.2	42.1	51.1	60.2	68.8	76.1	77.7	71.9	61.8	52.8	41.3
		MCDB	37.1	39.9	45.6	56.1	66.4	72.4	78.8	80.7	75.6	66.0	55.9	44.0
Mean Daily Temperature Range	5% DB	MDBR	9.1	8.9	9.1	10.4	10.3	9.4	8.0	8.5	10.1	10.5	10.1	9.2
		MCDBR	11.1	9.9	10.9	12.8	12.2	12.2	10.4	10.3	12.2	12.2	12.6	10.8
		MCWBR	10.7	8.2	8.2	8.4	7.6	7.7	6.3	7.0	8.5	9.4	10.4	10.3
	5% WB	MCDBR	10.3	9.1	10.7	12.3	10.9	10.5	8.8	8.7	11.2	10.6	10.6	10.0
		MCWBR	10.6	8.2	8.7	9.5	7.6	7.4	6.0	6.5	9.0	8.8	10.1	10.6
Clear-Sky Solar Irradiance	taub	0.429	0.496	0.547	0.620	0.648	0.735	0.715	0.615	0.541	0.565	0.516	0.443	
	taud	1.961	1.831	1.756	1.661	1.657	1.559	1.645	1.840	1.934	1.778	1.836	1.947	
	Ebn at Noon	218	220	224	218	215	197	200	217	224	196	186	201	
	Edh at Noon	42	55	66	77	78	87	79	63	54	57	48	40	
All-Sky Solar Radiation	RadAvg	772	1026	1403	1643	1807	1620	1443	1504	1428	1089	760	653	
	RadStd	58	80	104	125	150	182	149	162	112	71	77	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page